2019 International Conference on Computational Intelligence and Knowledge Economy (ICCIKE)
December 11–12, 2019, Amity University Dubai, UAE

Asset Locator with Distance and Altitude Measurement Using GPS and BLE for Future IoT Applications

Aquino, Mikaela L Costales
Electronics Engineering Department
Adamson University
Manila, Philippines
mikaela.aquino@adamson.edu.ph

Kimberly Rose B
Electronics Engineering Department
Adamson University
Manila, Philippines
kimberly.rose.costales@adamson.edu.ph

Araña, Jasper Meynard P.
Electronics Engineering Department
Adamson University
Manila, Philippines
jasper.arana@adamson.edu.ph

***Abstract*— Most people forget, misplace or lost valuables such as keys, wallets, documents, and even high value assets. Previous inventions relate in the field of locating assets, however having a wide coverage scope, hasn't been a priority. There is a need to develop a system to determine the exact location of objects, whether outdoor or indoor, and can calculate the distance and altitude from the user for reduction of time in searching for items. Integrating the locating of objects, whether outdoor or indoor, with distance and altitude measurement is advantageous because it promotes convenience which works in a way that it reduces time and effort of a person who is monitoring or searching for an item.**

The device developed will be able to assist an individual in locating misplaced or lost assets. The user will be able to save time and effort searching for misplaced or lost assets. Several tests were conducted to ensure the reliability and efficiency of the device in terms of displaying the pin in the map programmed in the mobile application. The device would uses GSM connectivity to give its location. All the tests are done to achieve the objectives of the study. These tests were able to produce positive results which made the researchers to conclude that the design project has been working properly in terms of its specifications and limitation.

Keywords—asset locator, distance, altitude, mobile application

I. Introduction

Valuables such as electronic devices, keys, and wallets are often forgotten, misplaced, or lost. The number of personal belongings that we misplace each day has no official figure, but one study claims that we spend at least 10 minutes per day looking for an item. There are a lot of reasons why people tend to lose things: forgetfulness is part of adult life for many reasons, including being preoccupied for a person with a hectic lifestyle or distracted by chores; a person is mentally overloaded, absent-minded or under stress; fatigue or health issues like depression, hypothyroidism or worse, dementia. However, everyday forgetfulness is quite common, and occasional memory lapse is normal at any age. Moreover, item may be difficult to locate because of its small size, it may be underneath other objects, or is in different location. We are familiar with the sense of panic and frustration when we realize that our valuable is missing especially when it's something we really want in a short period of time.

Many people struggle to keep track of their belongings because one forgets where he or she put the item or misplaced valuables like keys, wallet, documents, or mobilephones. Almost 3,000 individuals spend 10 minutes per day looking for misplaced items, but even if we only waste 10 minutes per day, that has an impact of which we may not be consciously aware [1]. These figures vary from year to year, country to country, and of course some items are misplaced over and over. People have either been late to work or school because of misplaced items, who have missed appointments or meetings. Unplanned search times and being delayed from continuing one's actions at work that can also harm your career [2]. The process of how to locate items is an issue that needs to be addressed in the most efficient way. This technology issue may be defeated through technologies designed to improve the user's experience in finding an item.

There has been ongoing development with respect to systems for tracking and locating assets. Many contemporary tracking systems rely on the Global Position System (GPS), a satellite-based navigation system. GPS receivers can calculate their position based on information transmitted by at least three of the approximately thirty GPS satellites in orbit about the earth. The use of GPS for obtaining location information is well-known; what is less apparent is the effective use of the location data to accomplish various goals [3]. Radio Frequency Identification is also utilized from past studies, which uses tags that only work within a short range making them ineffective for active tracking purposes and these signals are easily blocked by objects and other radio waves [4]; Moreover, Wi-Fi based system needs internet connection in order to work as tracker, however has a more substantial difference when range is stretched as a weaker signal is reflected in the connection; Visual and audio alerts are also available, however, it is difficult to trace when you go farther from the item. Previous inventions relate in the field of locating assets, however, a wide coverage scope hasn't been a priority. [5]-[7]

Integrating the locating of objects, whether outdoor or indoor, with distance and altitude measurement is advantageous because it promotes convenience which works in a way that it reduces time and effort of a person who is monitoring or searching for an item.

II. Project Design

A. Objectives

The main objective of the study is to develop an asset tag incorporated to a mobile application which will guide the user in locating items. This study specifically aims to develop a system that (a) notify the user of the location of an asset (b) reduce time in searching for assets (c) characterize tagging devices that are capable of pinpointing exact location of an asset, measuring the distance and altitude from an asset

978-1-5386-9346-9/19/$31.00 ©2019 IEEE

Authorized licensed use limited to: University of Exeter. Downloaded on May 06,2020 at 10:05:28 UTC from IEEE Xplore. Restrictions apply.

to the mobile phone and alerting the user with an audible sound system. Also, develop a program for (a) searching the position of each tag that the user wishes to locate (b) determines the estimated distance of the specified tag to the device (c) has a database of the asset tag to locate, and manage the logs. We will validate the device and system in terms of its reliability and efficiency.

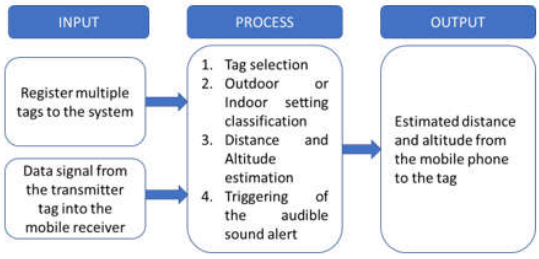

Fig. 1. Block Diagram of the Asset Locator

Figure 1 shows the overall process performed by the system. The system takes the multiple tag registration. Once a tag is already selected, the tag will transmit data signals into the receiver that will automatically determine which transmission technology to be used. The interface automatically displays information about the location of the selected tag.

B. Design

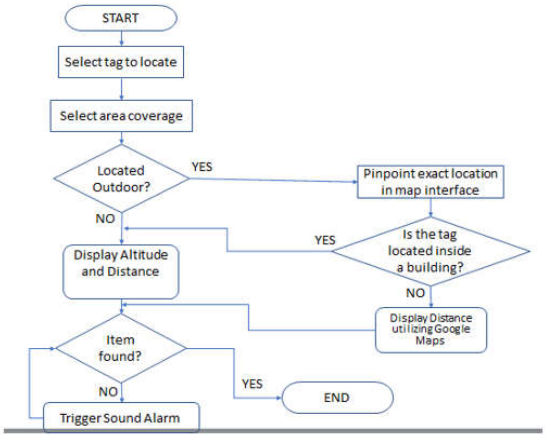

Fig 2. Flowchart of the Asset Locator System

Figure 2 shows the flow of the system where the step by step process is illustrated from input to end. At the start, the selection of tag and the expected setting are inputted by the user. Then, the exact location of the asset will be displayed in the map interface of the mobile application. The position of the desired tag will be identified whether outdoor or indoor setting. When located outdoor, only the distance will be calculated. When located indoor, distance and altitude will be displaced in the interface. If the asset is still not found, an audible sound alarm can be triggered to assist the user in searching for the item. The sequence shall repeat if the item is still not found. The system can repeat the process since it can accommodate multiple tags.

Figure 3 shows the detailed design of the device. It is composed of GPS and GSM module, Bluetooth Module, an Arduino as its microcontroller, an audio module, a speaker switch and battery for the tag to function.

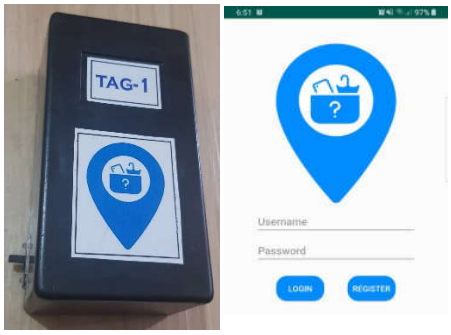

Fig. 3. Design Layout of the Prototype and Mobile Application

III. Results and Discussion

Data and results gathered were based on the criteria set by the researchers. It was tested for its reliability and efficiency on pinpointing the exact position of the asset. There were three main criterion for this, the GPS measurement in terms of latitude and longitude, distance of asset to the mobile device, and the altitude measurement.

A. GPS

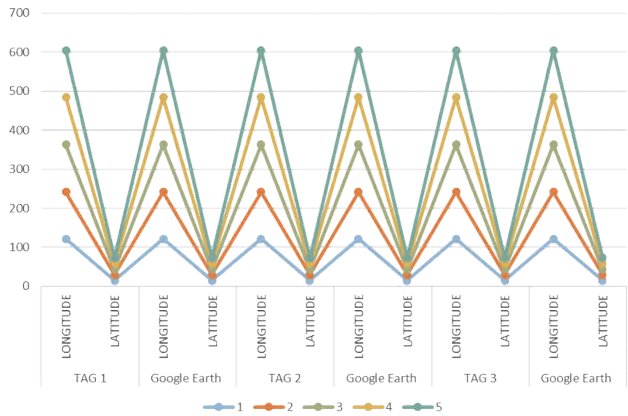

Fig 4. Comparison of reading for both longitude and latitude for Tag 1, 2 and 3

The data gathered by the GPS was compared to Google Earth to test the reliability of the said data. Figure 4 shows a sample results of the reading of both longitude and latitude for tag 1, 2, and 3. The coordinates were expressed in decimal degrees. The researchers performed the test in 5 different locations. In each locations 30 trials were performed to test the response of the system.

The average value for longitude for tag 1, 2 and 3, independently was 121.15 with a standard deviation of 0.66 and for latitude average of 14.25 with standard deviation of 1.11. For the Google earth reading, tag 1, 2 and 3, independently was 121.15 with a standard deviation of 0.66 and for latitude average of 14.25 with standard deviation of 1.11. Computed p-value for longitude based on paired t-test was 0.1724 which implies that there is no significant difference between the reading of the locator asset device and Google earth coordinates.

Authorized licensed use limited to: University of Exeter. Downloaded on May 06,2020 at 10:05:28 UTC from IEEE Xplore. Restrictions apply.

Computed p-value for latitude based for on paired t-test was 0.2736 which implies that there is no significant difference between the reading of the locator asset device and Google earth coordinates. Therefore, the location obtained from the asset tag is consistent in displaying the exact position of an item.

B. Distance

TABLE 1 RELIABILITY OF THE ASSET LOCATOR (DISTANCE)

TRIAL	MANUAL (m)	TAG1 (m)	MANUAL (m)	TAG2 (m)	MANUAL (m)	TAG3 (m)
1	26	25.982	63	62.982	46	45.992
2	44	43.964	42	41.985	80	ERROR
3	79	ERROR	68	68.063	67	67.065
4	65	64.947	34	34.006	55	55.035
5	53	53.059	27	27.064	90	ERROR
6	41	41.011	19	19.023	75	75.057
7	57	57.054	82	ERROR	68	67.992
8	36	35.973	45	45.044	15	15.002
9	25	25.018	52	51.983	30	30.008
10	46	46.064	38	38.012	40	40.054

Table 1 shows the data for measurement of the distance. An observer is on one place outside the building with the mobile phone. The testing is done simultaneously by measuring the displayed altitude and distance; all the tags are located at different locations inside the building. The manual distance is measured using the instrument, Tape Measure Laser 196. The measurements in two-dimensions, the x and y axis, are used. The table also shows that the system may display errors because it cannot compute for the distance. The system is limited only to 77 meters below with respect to the x-axis because of the specification of the used Bluetooth module. Measurements above the specifications cannot be measured and will display error.

C. Altitude Measurement

TABLE 2. HEIGHT MEASUREMENT OF SUNTRUST CONDOMINIUM

TRIAL	*MANUAL (m)*	*TAG1 (m)*	*MANUAL (m)*	*TAG2 (m)*	*MANUAL (m)*	*TAG3 (m)*
1	3	2.9932	6	5.9885	9	9.0021
2	9	8.9963	12	11.9813	6	5.9943
3	12	12.0007	3	3.004	3	3.001
4	6	6.0030	9	8.9992	6	5.9864
5	9	8.9996	9	8.9869	9	9.0031
6	12	12.0003	3	2.9831	3	2.9876
7	6	5.9453	6	5.9964	6	6.0012
8	3	2.9846	12	11.9914	12	12.0001
9	6	5.9912	3	2.9816	3	2.9974
10	6	6.0021	6	6.0027	12	11.9978

Table 2 shows the data for the altitude measurement. Data were gathered by an observer using the mobile application located outside the building. The testing was done simultaneously by measuring the displayed altitude and distance, with all the tags positioned at different locations inside the building.

The National Building Code of the Philippines, Chapter 1, Section 104 states that the standard height of a building is approximately 9 feet (3 meters) inclusive of the thickness of the flooring [8]. The manual height of the building is measured using the instrument, Tape Measure Laser 196. The measurements are in two-dimensional, only the x and y axis are used.

IV. SUMMARY OF FINDINGS

After using different statistical tool and techniques for data analysis, the summary of findings is the following:

1.The low-powered asset-locating device can pinpoint and exact location, measure the distance and altitude of an asset to the mobile phone.

2.The mobile application has a data base and interface that can display the asset's location by coordinates on the map. It can also display the estimated measurements of the distance and altitude.

3.The device is reliable for the system is consistent in locating the object and displaying the distance and altitude.

4.The device is efficient for it helped the user reduce the time in searching for an asset.

V. CONCLUSION

The researchers were able to design and develop an asset-locating device incorporated to a mobile application that can assist an individual in locating misplaced or lost items and will be able to reduce time and effort searching for items.

REFERENCES

[1] "*What did you lose today?"* Accessed on: December 2018. [Online]. Available: http://www.myntracker.com/mynt

[2] "*Measuring a Distance between Things with Improved Accuracy*." Accessed on: December 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1877050915009199

[3] M. Leeslie, J. Reilander, U.S. Publication No. 2013/0342394 A1*, Apparatus and System for Tracking Packaging Including Bait Packaging*

[4] "Technovelygy.com. *Technical problems with RFID"* Accessed on: November 2018. [Online]. Available at: http://www.technovelgy.com/ct/Technology-Article.asp?ArtNum=20. [5] E. Young, U.S. Publication No. 2014/0266756 A1, *Object Tracking with Platform Notifications*

[5] Y.S. Pan, K.H. Chong, WO 2014/042507, *Wireless Pairing and/or Tracking System for Locating Lost Items*

[6] Shoari and Komeijani, US 10,186,139 B2, *Apparatus and methods to promote sustainable behavior among smartphone users*

[7] *National Building Code of the Philippines, Presidential Decree No. 1096,* Malacanang, Manila, Philippines 2005

Authorized licensed use limited to: University of Exeter. Downloaded on May 06,2020 at 10:05:28 UTC from IEEE Xplore. Restrictions apply.